IT41033 - Nature Inspired Algorithms

MINI PROJECT REPORT

Lightweight Machine Learning Model for Early Diabetes Prediction Using Clinical Data



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Abstract

Diabetes is a problem for public health and it is getting worse. So, it is an idea to have a simple way to check for early signs of diabetes. This project used a machine learning process to try to predict if someone has diabetes. The project used a dataset that had information about 768 patients. This dataset had eight health numbers and one label that said if the patient had diabetes or not. When the project team looked at the data, they found that some of the numbers were not correct. For example some patients had glucose levels of zero, which's not possible. So, the team changed these numbers to say that the information was missing. Then they split the data into two groups: one group to train the machine learning model and one group to test it. The team used a way to fill in the missing numbers and to make all the numbers similar. The team found that some of the health numbers were more important than others for predicting diabetes. They used a method to choose the most important numbers. Then they tried machine learning models to see which one was the best. They compared the models by looking at how accurate they were and how well they could predict diabetes. The team found that one of the models called Logistic Regression was the best. This model was very small. Could make predictions very quickly. It was able to predict diabetes about 72 percent of the time. The team was happy with the results. They know that the model needs to be tested in a real world setting before it can be used. Diabetes is a disease and any model that tries to predict needs to be very accurate. The model is very small. Can be stored on a computer using very little space. It can also make predictions quickly. The team thinks that this model could be useful for screening patients for diabetes. They need to do more testing to make sure it is accurate and reliable. The team hopes that their model can help doctors and patients catch diabetes early and prevent health problems. They think that their model is a start, but it needs more work before it can be used in real life.

Keywords - diabetes prediction, lightweight machine learning, Logistic Regression, feature selection, clinical screening, model compression.

1. Introduction

1.1 Background and motivation

Diabetes mellitus is a metabolic disorder that is characterized by several severe complications affecting the cardiovascular system, kidney, nervous system and vision. According to the International Diabetes Federation, there were 589 million adults aged 20-79 years old suffering from diabetes in 2024, and the number is expected to reach 853 million by 2050 [1]. Such statistics serve to emphasize the need for cheap screening methods able to select high-risk patients and refer them for further diagnosis.

Such routine measures used in clinic practices as plasma glucose level, body mass index (BMI), age and family history criteria are easily available and convenient for structured data classification. Goodrtheless, good discriminatory ability alone does not ensure the feasibility of the screening test in low-resource settings. To be effective, a screening model must be compact, cheap in computation, reproducible and clear about its uncertainties. Hence, the current research aims to evaluate the performance of classical machine learning models.

1.2 Problem statement

Many studies on predicting diabetes focus on maximum accuracy at the expense of a complicated structure of the model, which may consume much memory and computational resources that may not be available in small clinics, mobile applications or edge devices. On the other hand, a very simple model may miss patients who need to be followed up. The challenge here is to find a few useful clinical factors and a small but effective classifier.

1.3 Research Questions

- RQ1: Which subset of the eight PIMA clinical attributes provides the strongest predictive value while minimizing the number of required inputs?
- RQ2: Can Logistic Regression, Decision Tree and Gaussian Naive Bayes models trained on the selected subset approach or exceed 80% held-out accuracy and remain competitive with a Random Forest baseline?
- RQ3: What model size and inference time can be achieved through lightweight optimization while keeping any accuracy decrease below 3%?

1.4 Objectives

- Prepare the PIMA dataset by identifying invalid zeros, imputing missing measurements, scaling features and creating a stratified 80:20 split.
- Use correlation analysis and recursive feature elimination (RFE) to select a compact feature subset.
- Train and compare three lightweight classifiers against a Random Forest baseline.
- Evaluate models using accuracy, precision, recall, F1-score, ROC-AUC and five-fold cross-validation.
- Tune model hyperparameters, measure storage and latency, and compress the preferred model.

1.5 Scope and contribution

The applied study relies on one data source - Primary PIMA Indians Diabetes Dataset. The CDC Diabetes Health Indicators data source mentioned in the proposal has not been utilized in the final experiment, while there have been no local sources of clinical data. The project contribution consists in the leakage-sensitive comparison of the data quality, feature reduction, model validation and deployment effectiveness. The project result is a prototype for risk screening, not a diagnostic tool.

2. Literature Review

2.1 Diabetes risk prediction from structured clinical data

The dataset PIMA is well known for its use in the analysis by Smith et al., where an adaptive learning approach was applied to predict the development of diabetes mellitus [2]. The small sample size of the PIMA database and the combination of the binary variable with a few clinically meaningful predictors make it a good educational example despite its limited external validity.

A frequently occurring problem is that there are zeros in some clinical variables that have no physiological meaning. If zeros are considered true measurements, they can affect descriptive statistics and modeling borders. In the current study, zeros in the variables Glucose, BloodPressure, SkinThickness, Insulin and BMI were regarded as missing and imputed based on medians estimated exclusively from the training subset. The reason for such an imputation strategy is that many variables have skewed distributions and extreme values.

2.2 Candidate classification algorithms

Logistic Regression calculates the probability of using a linear decision function and is appealing from the deployment perspective, as it requires saving only a handful of coefficients. It also allows regularization and class weight. [3] Decision Trees represent non-linear decision boundaries and have direct interpretability, but uncontrolled trees are prone to high variance and overfitting. Random Forest is used to overcome the variance problem in the Decision Tree by averaging several trees trained randomly [3] it usually gives better predictions at the cost of a much bigger size of a serialized model and slow inference speed.

Gaussian Naive Bayes uses the Bayes' theorem with class conditional features modeled by Gaussians and assuming conditional independence between features. Despite the simplicity of the latter assumption, the approach is cheap computationally and can work well on numerically small data. [5]

2.3 Feature selection and reliable evaluation

Feature selection can help reduce cost of acquisition, storage and computing and also remove weak predictors. The recursive feature elimination technique iteratively fits an estimator and eliminates the least important feature until only a specified subset of features is left [6]. Since preprocessing and selection techniques have the potential to leak information from validation folds, they need to be fit within the training process instead of being trained on the entire dataset. Scikit-learn offers pipelines, RFE, cross-validation and grid search techniques [4].

Precision on its own could be misleading when it comes to imbalanced classes. Precision is the proportion of positive predictions which are correct, recall is the proportion of positive cases which have been identified and the F1-score combines both precision and recall into account. The ROC-AUC score looks at ranking discrimination across all thresholds [5]. When we talk about screening, recall becomes especially important due to false negatives delaying referral; however, false positives are undesirable too.

2.4 Lightweight deployment

A lightweight model should be assessed as a complete inference pipeline, including imputation, scaling and feature selection where applicable. Reducing numeric precision from Float64 to Float32 can lower storage, while general-purpose compression can remove serialization redundancy. Such optimization is acceptable only when performance is checked after conversion. This study measures model size, batch latency, per-patient latency and any change in held-out accuracy and ROC-AUC.

3. Methodology

3.1 Research design and tools

The project was carried out according to a quantitative experimental process that involved problem identification, data understanding, preprocessing, feature reduction, modeling, model validation, model tuning, resource benchmarking and model interpretation. The tools used were Python in Google Colab along with packages like pandas, NumPy, Matplotlib, Seaborn, scikit-learn and joblib. Scikit-learn is a machine learning tool written in Python for data mining and data analysis tasks [4].

3.2 Dataset and variables

There was a total of 768 observations with 9 numeric variables. The outcome variable was binary with "0" for no diabetes and "1" for diabetes. A summary of predictors is shown in Table 1.

Attribute	Meaning	Role
pregnancies	Number of Pregnancies	Predictor
Glucose	Plasma glucose concentration	Predictor
BloodPressure	Diastolic blood pressure (mm Hg)	Predictor
SkinThickness	Triceps skin fold thickness (mm)	Predictor
Insulin	Two hour insulin serum	Predictor
BMI	Body mass index	Predictor
DiabetesPedigreeFunction	Family history risk function	Predictor
Age	Age in years	Predictor
Outcome	0 = no diabetes, 1= diabetes	Target

Table 1: Dataset attributes used in the study.

3.3 Exploratory data analysis

The analysis examined the shape, data type, descriptive statistics, missing data, duplicates, class distribution, histogram, correlation, and box plots. The conventional NaNs and complete duplications were

not detected. However, clinical zeros had to be handled separately. There were 652 clinically erroneous zeros found, mainly in the columns of Insulin and SkinThickness.

Table 2: Clinically invalid zero measurements detected before preprocessing

Clinical Attribute	Invalid Zeros
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11
Total	652

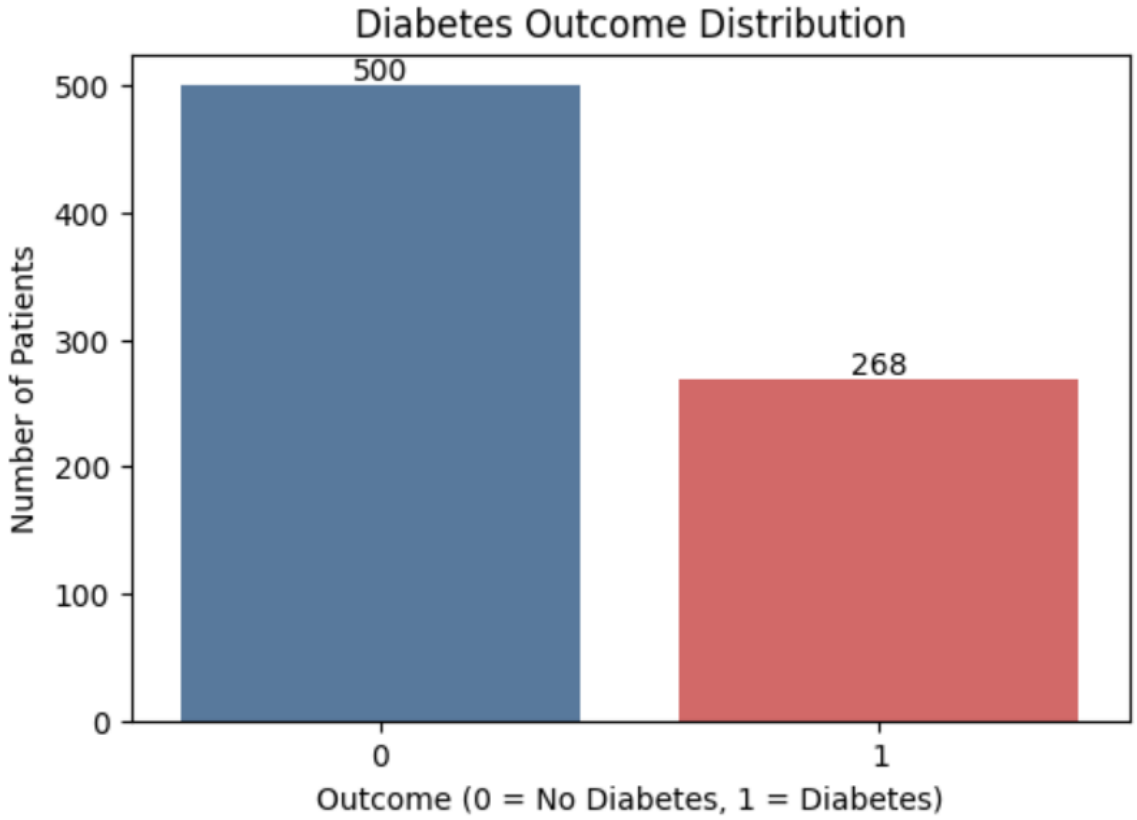


Figure 1: Outcome class distribution before modeling.

Imbalance existed in the target distribution in which 500 instances (65.1%) were in class 0 while 268 instances (34.9%) were in class 1. Glucose possessed the highest correlation with Outcome (0.47),

followed by BMI (0.29), Age (0.24), Pregnancies (0.22) and DiabetesPedigreeFunction (0.17). Correlation is only descriptive but not necessarily adequate.

3.4 Preprocessing and leakage control

All invalid zeros in the five problematic clinical columns were marked as NaN. Then predictors and target variables were separated and stratified train-test split was conducted with 614 and 154 observations for training and testing respectively. In the training dataset the proportions of classes are 65.15% and 34.85% for class 0 and 1 respectively in the testing dataset the proportions are 64.94% and 35.06% correspondingly.

Median imputation and standardization were performed on the training set only and then used for the test set. Medians of the main training variables are Glucose - 117, BloodPressure - 72, SkinThickness - 29, Insulin - 125 and BMI - 32.4. After median imputation there are no missing values left. Features standardized in the training set have mean around zero and standard deviations around one.

3.5 Feature Selection

The RFE algorithm using logistic regression as an estimator chose five out of eight variables. This same set of selected variables was chosen by the three light-weighted algorithms [9]. However, the random forest baseline kept all eight variables as proposed initially.

Feature	Selected	RFE Rank
pregnancies	Yes	1
Glucose	Yes	1
BloodPressure	Yes	1
DiabetesPedigreeFunction	Yes	1
Age	Yes	1
Insulin	Yes	2
BMI	Yes	3
SkinThickness	Yes	4

Table 3: RFE feature-selection result.

3.6 Model development and evaluation

First, four baseline classifiers were trained: Logistic Regression, Decision Tree, Gaussian Naive Bayes and a 100-tree Random Forest. Predictions and probabilities were made on the held out test dataset. Confusion matrices were used for counting true negatives, false positives, false negatives and true positives. After this, five-fold stratified cross validation was used to measure average performance and fold stability. Preprocessing and RFE was placed in the model pipeline during cross validation to allow each fold to learn transformations from its own training dataset.

GridSearchCV was used with five-fold stratification and optimization based on ROC-AUC score to find best hyperparameters. The grid search for the Logistic Regression included regularization strength, penalty type, solver and class weighting, grid search for the Decision Tree included controls related to pruning, Gaussian Naive Bayes used variance smoothing and Random Forest used tree depth, feature subsampling, minimum leaf size and class weighting [10].

3.7 Resource optimization and final demonstration

The size of the serialized pipeline was kilobytes. The inference time was calculated using the 154-record test dataset and divided by the batch size to get an estimation of latency per patient. The chosen Logistic Regression pipeline was then downcast from Float64 to Float32 and saved in joblib format. The accuracy and ROC-AUC score was recalculated after the optimization process. Lastly, one patient's record was fed into the entire pipeline to illustrate the classification and probability output.

4.Results

4.1 Baseline model performance

Model	Train Acc.	Test Acc.	Precision	Recall	F1 - Score	ROC - AUC
Random Forest	1000	0.7662	0.6957	0.5926	0.6400	0.8210
Gaussian Naive Bayes	0.7834	0.7338	0.6226	0.6111	0.6168	0.8017

Logistic Regression	0.7964	0.7013	0.5870	0.5000	0.5400	0.8102
Decision Tree	1000	0.7013	0.5800	0.5370	0.5577	0.6635

Table 4: Initial held-out test performance.

Random Forest had the starting test accuracy and ROC–AUC but both Random Forest and Decision Tree reached perfect training accuracy, which suggests overfitting. The initial confusion matrices were as follows: Logistic Regression had 81 TN, 19 FP, 27 FN and 27 TP; Decision Tree had 79 TN, 21 FP, 25 FN and 29 TP; Gaussian Naive Bayes had 80 TN, 20 FP, 21 FN and 33 TP; and Random Forest had 86 TN, 14 FP, 22 FN and 32 TP.

4.2 Cross validation results

Model	CV acc.	CV Precision	CV Recall	CV F1	CV AUC	Acc. SD
Logistic Regression	0.7850	o.7552	0.5748	0.6512	0.8445	0.0189
Gaussian Naive Bayes	0.7773	0.7188	0.5937	0.6482	0.8427	0.0230
Random Forest	0.7606	0.6899	0.5750	0.6258	0.8163	0.0188
Decision Tree	0.6499	0.4974	0.5467	0.5200	0.6259	0.0343

Table 5: Five-fold stratified cross-validation performance.

The results after cross-validation differed from those of the initial test set division. The best mean accuracy, F1-Score, and ROC–AUC were obtained for Logistic Regression - 0.7850, 0.6512, and 0.8445 correspondingly, with minimal accuracy variance. Gaussian Naive Bayes performed similarly well. The decision tree without any limitations was the most unstable algorithm.

4.3 Tuned model results

Model	Train acc.	Test acc.	Precision	Recall	F1	Test AUC	Best CV AUC
Logistic Regression	0.7524	0.7273	0.5882	0.7407	0.6557	0.8128	0.8483
Gaussian Naive Bayes	0.7834	0.7338	0.6226	0.6111	0.6168	0.8017	0.8427
Random Forest	0.8371	0.7338	0.6585	0.5000	0.5684	0.8031	0.8422
Decision Tree	0.7280	0.6818	0.5287	0.8519	0.6525	0.7840	0.7963

Table 6: Final tuned-model comparison.

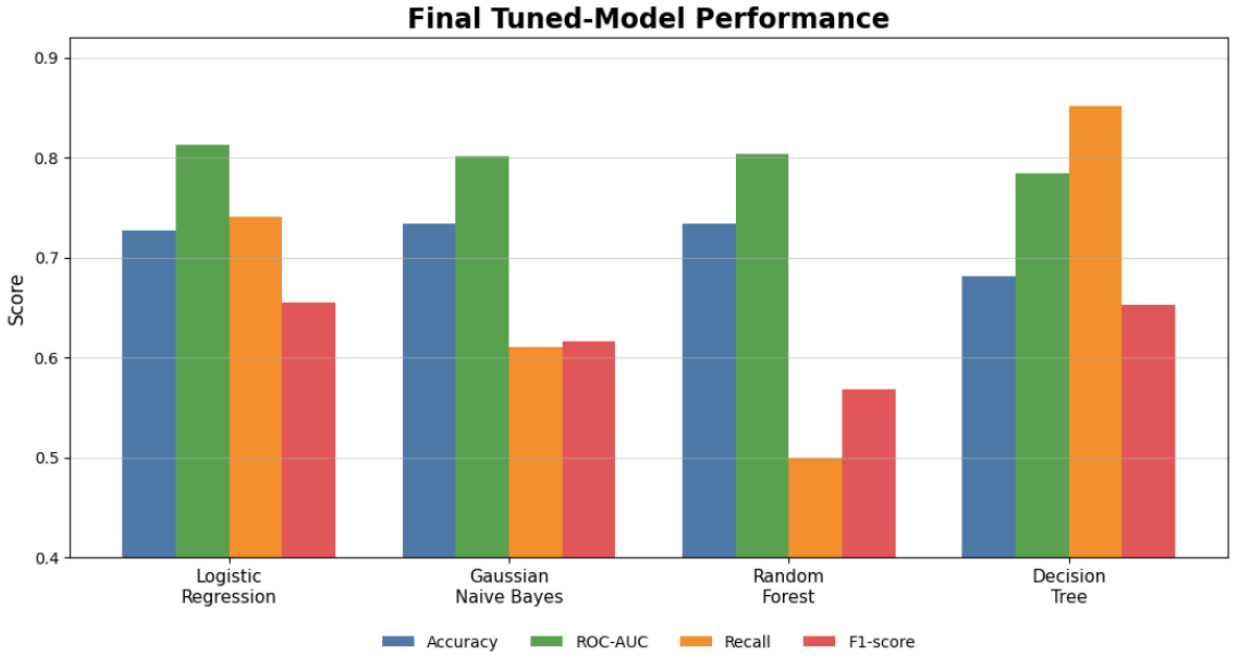


Figure 2: Comparison of the four tuned models on the held-out test set.

The Logistic Regression model we picked had a penalty called L2 regularization. This penalty had a value of C which was 0.01. We also used the solver and made sure the class weights were balanced. The model only used 5 features that were chosen by a process called RFE. This model did well. It got the score when we tested it with something called cross-validation ROC-AUC. The score was 0.8483. It also did well when we looked at the test F1-score. That score was 0.6557.When we adjusted the model to make it better it got even better at finding the things. It went from 0.5000 to 0.7407. It was not very good at being precise.We also tried to make a Decision Tree model. The best tree had a depth of 3 and 8 leaf nodes. It

was really good at finding things too. Its recall was 0.8519. It was not as good at getting the right answer all the time. It also did not do well when we looked at the ROC-AUC values.The Gaussian Naive Bayes model and the Random Forest model did well when we looked at how accurate they were. The Gaussian Naive Bayes model was not as good at finding the right things or getting a good F1-score, as the Logistic Regression model.

4.4 Model size and inference time

Model	Size (KB)	Batch time (ms)	Time / Patient (ms)
Logistic Regression	3.1064	2.7531	0.0179
Gaussian Naive Bayes	3.3369	2.8859	0.0187
Decision Tree	4.8545	2.7931	0.0181
Random Forest	356.4082	29.7571	0.1932

Table 7: Serialized pipeline size and inference time.

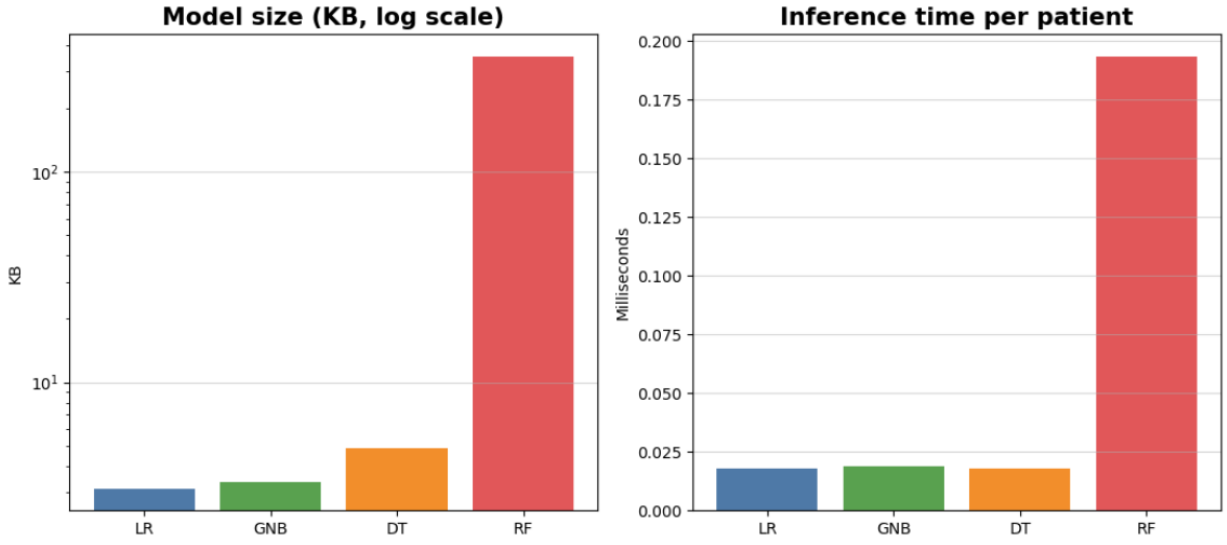


Figure 3: Resource comparison; the model-size axis is logarithmic.

Logistic Regression had the least size and speed of execution. The Random Forest model needs almost 115 times more space and nearly 10.8 times more inference time per patient compared to Logistic Regression. This is environment-dependent, but the scale of difference justifies choosing a light model.

4.5 Quantization, compression and sample prediction

Version	Size (KB)	Accuracy	ROC-AUC	Time /Patient (ms)
Original flot32	3.1064	0.7273	0.8128	0.0179
Quantized flot32	2.9658	0.7273	0.8128	0.0200
Flot32 + compression	1.5176	0.7273	0.8128	0.0200

Table 8: Optimization of the selected Logistic Regression pipeline.

The Float32 compression and conversion led to the file size decreasing from 3.1064 KB to 1.5176 KB, for a 51.1% reduction in file size. The held-out accuracy and ROC-AUC were not affected by the changes, and thus, the accuracy difference was 0%, which meets the condition stated by the proposal, that the reduction in accuracy should be less than 3%.

The final result of the pipeline for the demonstration patient, who has 6 pregnancies, a glucose level of 148, blood pressure of 72, skin thickness of 35, insulin value recorded as 0, BMI of 33.6, Diabetes Pedigree Function 0.627 and age 50, indicates that the class is 1 and diabetes probability is 71.39%.

5. Discussion

5.1 Interpretation of findings

RQ1 was answered by a five-feature subset, including Pregnancies, Glucose, BMI, DiabetesPedigreeFunction and Age. Glucose turned out to have the highest correlation with Outcome variable among all features, while BMI, age, pregnancy history and pedigree data provided additional risk-related information. Exclusion of BloodPressure, SkinThickness and Insulin from the inputs decreased the input dimensionality and contributed to the construction of a more compact model. This selection must be treated as being relevant to the specific dataset rather than a clinical assertion that the excluded measurements are irrelevant across the board.

As to RQ2, all models failed to meet the 80% accuracy goal on the held-out dataset. The highest tuned held-out accuracy was 73.38%, which was achieved by Gaussian Naive Bayes and Random Forest. Mean five-fold accuracy, however, showed that Logistic Regression was 78.50% accurate, thus approaching the target value and varying depending on sample splitting. Logistic Regression proved to provide the best compromise between the requirements of achieving good performance and saving resources: the highest

tuned cross-validation ROC-AUC score, highest tuned F1-score, significantly increased recall and minimum resource consumption. Therefore, the accuracy goal was partially achieved, while the efficiency and comparative goals were met.

Training perfect accuracy of Decision Tree and Random Forest was a clear sign of overfitting. Decision Tree depth was reduced from 15 to 3, while training accuracy dropped from 1.0000 to 0.7280, and recall increased to 0.8519. This result is very helpful for aggressive screening threshold, though poor ROC-AUC and accuracy indicate that there will be a lot of false-positive predictions. Tuning of Random Forest improved the training-test gap difference, but required much more resources than any other method.

5.2 Screening trade offs

The calibrated Logistic Regression algorithm correctly identified 74.07% of the positive test instances, an impressive feat compared to the uncalibrated model recall of 50.00%. The weighted balance towards the positive class might have changed the decision boundary to detect more positive samples. While lowering the chances of missing a case, it also led to the precision of only 58.82%, which implies that many positive flags did not represent positive cases. When designing the referral process, the thresholds of the model must be chosen with clinicians based on the cost considerations.

5.3 Lightweight objectives

Question RQ3 was positively answered. The pipeline, after being optimized, used 1.5176 KB while preserving its accuracy and ROC - AUC. The speed at which the pipeline makes predictions, approximately 0.0200 ms per patient, is sufficient for fast processing. This means that model compression fulfilled the accuracy loss criterion for the project, which was less than 3%. However, benchmarking the entire pipeline is much more meaningful than just the classifier.

5.4 Limitation and threats to validity

- The dataset consists of only 768 records and is specific to the population in question; therefore, findings may not extrapolate to patients in Sri Lanka, males, any other age group, or other settings.
- The Outcome attribute and past data collection procedure may not adhere to the present-day criteria and procedures for diagnosis and screening.

- Just one 20% test set consists of merely 154 records. Cross-validation minimizes but does not completely eradicate variance.
- Median imputation is efficient yet reliable, although it tends to obscure clinically relevant missing data, particularly in cases of Insulin and SkinThickness.
- There was no external dataset provided. The CDC dataset and additional local validation datasets have not been used within the project's scope.
- The prediction is neither calibrated nor clinically verified, and performance was not analyzed for various demographic subpopulations.
- Latency metrics were recorded from a Colab notebook using a small batch and require repetition in the target setting.

5.5 Recommended future work

Further work should include validation of the entire pipeline on a bigger and more locally relevant dataset, assessment of probability calibration and thresholds, confidence intervals, subgroup comparison, and analysis of the impact on clinical workflow. Nested cross validation will give a more conservative measure where both model selection and model evaluation are computationally demanding. Should appropriate data become available, recalibration or retraining of the model on Sri Lankan clinical data would be required before conducting any deployment study.

6. Conclusion

This project designed and implemented a full-lightweight diabetes prediction pipeline with the PIMA Indians Diabetes dataset. There is moderate class imbalance and 652 clinically invalid measurements with zero value found during data exploration. The leakage-free preprocessing pipeline included training-only median imputation and scaling as well as RFE for reducing number of predictors from eight to five (Pregnancies, Glucose, BMI, DiabetesPedigreeFunction and Age).

Four classification algorithms were tested by held-out performance, confusion matrixes, five-fold cross-validation and hyperparameter tuning by ROC-AUC criterion. The tuned Logistic Regression was chosen as the algorithm which provides the optimal combination of discrimination, recall, F1-score, stability and speed. It achieved the accuracy of 72.73% on held-out data, 74.07% recall rate, F1-score of 65.57%, test ROC-AUC of 0.8128 and cross validation ROC-AUC of 0.8483. The full pipeline including Float32

conversion and compression occupied only 1.5176 KB of space without any loss of accuracy and ROC-AUC.

The goal of 80% held-out accuracy wasn't reached, so this project doesn't provide clinical-grade prediction ability. However, it proved that a small classical machine learning model can produce decent discrimination with extremely little storage requirements and latency. The final pipeline should be considered as an educational prototype for further independent clinical validation, local population estimation, calibration and review of the model's usage governance policy.

References

[1] I. D. Federation, "IDF Diabetes Atlas, 11th edition," 2025. [Online]. Available: https://diabetesatlas.org/resources/idf-diabetes-atlas-2025/. [Accessed 15 August 2026].

[2] J. W. Smith, J. E. Everhart, W. C. Dickson, W. C. Knowler and R. S. Johannes, "Using the ADAP learning algorithm to forecast the onset of diabetes mellitus - Proceedings of the Annual Symposium on Computer Application in Medical Care," 1988.

[3] J. R. Quinlan, "Induction of decision trees," *Machine Learning,* no. 1, pp. 81-106, 1986.

[4] L. Breiman, "Random forests," *Machine Learning,* vol. 45, no. 1, pp. 5-32, 2001.

[5] G. H. John and P. Langley, "Estimating continuous distributions in Bayesian classifiers - Proceedings of the 11th Conference on Uncertainty in Artificial Intelligence," 1995.

[6] I. Guyon, J. Weston, S. Barnhill and V. Vapnik, "Gene selection for cancer classification using support vector machines," *Machine Learning,* vol. 46, pp. 389-422, 2002.

[7] F. Pedregosa, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research,* vol. 12, pp. 2825-2830, 2011.

[8] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters,* vol. 27, no. 8, pp. 861-874, 2006.

[9] s.-l. developers, "Recursive feature elimination," 2026. [Online]. Available: https://scikit-learn.org/stable/modules/feature_selection.html#rfe. [Accessed 15 August 2026].

[10] s.-l. developers, "GridSearchCV," 2026. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html. [Accessed 15 August 2026].

Appendix A. and reproducibility summary

Item	Setting
Environment	Google Colab / Python
Dataset size	768 records; 8 predictors; 1 binary target
Split	80% training / 20% testing; stratified; random_state = 42
Missing-data method	Invalid clinical zeros → NaN; training-fitted median imputation
Scaling	Training-fitted StandardScaler
Feature selection	RFE; five selected features
Validation	Five-fold stratified cross-validation
Tuning	GridSearchCV; ROC–AUC scoring
Recommended model	Balanced L2 Logistic Regression, C = 0.01, liblinear
Optimized artifact	Float32 plus joblib compression; 1.5176 KB